Inhaling Alcohol?
Vapshot is a new device that vaporises alcohol so that you can inhale it. Check it out: **http://dgit.in/Vapshot**

Solar powered smart benches
Soofa, a solar-powered smart bench lets you plug in and charge your phones and also connect to the internet using Wi-Fi **http://dgit.in/SmartBench**

istrar General and Census Commissioner found that 74 percent of slum dwellers have access to drinking tap water compared to 70.6 percent in urban areas! That said, 18.9 percent of slum dwellers still have no access to public or private toilet facilities and so, resort to defecating in the open.

Technology to the rescue!

While we aren't exactly on track to achieving the Millennium Development Goal of halving the population without access to drinking water and sanitation by 2015, we must do what we can to ensure that solutions are sustainable while also being cost-friendly. Technology provides such solutions in all cost ranges but let's look at the affordable options.

1. Bob, the rainwater storage bag

We can't talk about affordable options and leave out rainwater harvesting. One of the major obstacles to large-scale uptake of household level rain catchment is the high cost of storage and the difficulty of transporting large tanks to remote locations. Bob is an award-winning flexible 1,400 litre tank weighing only 3.5 kg . It has a built-in

A rural woman receiving Bob

filter and can fold up into a package the size of a large briefcase. The rainwater bag – purchasable from the open market in East Africa for $54/₹3282 (59 percent less than competing products) – is the brainchild of Joern Lutat, a Germany-based Mechanical Engineering college lecturer.

This solution will obviously only hold water, so to speak, in areas like Uganda and Ghana where drought and moderate to heavy rainfall are not impediments.

2. SlingShot Water Purifier

Created by Bionic Arm and Segway inventor, Dean Kamen, SlingShot is a water vapour compression distillation system. The system boils and evaporates any dirty water (yes, even sewage water), and then allows the pure water to condense and be collected. It can make up to 1,000 litres of water drinkable per day and requires as little power as required to work a toaster. It can be powered even by a generator running on cow dung. Notably, besides cleaning water, the Slingshot also generates small amounts of electricity – enough to light up a small village's energy-efficient light bulbs.

At the 2012 Clinton Global Initiative, Coca-Cola announced a partnership with Dean Kamen's company DEKA R&D to deliver Slingshot water purification systems to schools, community centres and health clinics in rural communities throughout countries such as Africa, Asia and Latin America. Each machine costing around $2,000 (₹1,21,580) will be deployed across 20 countries to serve 45,000 people by 2015. Coke won't directly profit from the program but will employ and train local entrepreneurs to operate what it's currently calling 'Ekocentres' – web-connected super-market centres that will be dropped into targeted underdeveloped villages to provide basic necessities. Each red Ekocentre will house canned food, toilet paper, cooking oil and first-aid supplies. Solar panels on the roof will power wireless Internet and a television. The SlingShot, about the size of a mini fridge, will be built into the centre with two faucets dispensing free purified drinking water to villagers.

3. LifeStraw

Designed in Switzerland by Vestergaard Frandsen, LifeStraw is a device that promises to remove 99.9999 percent of waterborne bacteria through a super-fine filtration process. Each LifeStraw, suitable for one person, costs only $5 (₹300) to manufacture and can make surface water safe to drink for up to a year. Over a period of three years from the date of manufacture, the LifeStraw filters over 1,000 litres – a limit you know you've reached when water can no longer be sucked through it. Since no replacement parts are needed, there's no additional cost.

All you do is suck water through the straw and waterborne protozoan parasites are cut down to 0.2 microns in size, providing five times more filtration than the US Environmental Protection (EPA) standard considered "rigorous" at 1.0 microns for removing Giardia, E. Coli and Cryptosporidium. The award-winning point-of-use LifeStraw technology is not just a lifesaver for rural dwellers but also for urbanites when there's a boil-water advisory combined with a power outage or if you're trekking in the middle of nowhere and critically need access to clean water.

While the LifeStraw removes nearly all bacteria and protozoa, it should be noted that it can't filter out heavy metals and viruses or desalinate water.

The bodega style centre will serve small villages with basic supplies

4. Waterchip

This is where Kyle Knust's Waterchip comes in. Researchers at the University of Texas at Austin – including Kyle Knust, a 27-year-old Analytical Chemistry student – are working on a plastic water chip which has a micro-channel with two branches. By applying a 3.0 volt electrical charge, they create an "ion depletion zone" with an embedded electrode that neutralises chloride ions. Doing this, they're able to redirect the salts in the water down one channel, while the fresh water goes down another. According to the team, "like a troll at the foot of the bridge, the ion depletion zone prevents salt from passing through, resulting in the production of freshwater".

The Waterchip, small enough to fit in the palm of your hand, is still having its kinks worked out. While it can also remove harmful bacteria, heavy metals and parasites, the chip can currently only remove a quarter (25 percent) of the salt from seawater, clearly not enough to produce water fit for drinking. However, this is an important invention to look forward to given how quickly climate change is slowly making freshwater rare to find.